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Cyber Security Internship

Report

WK1-PM4.b — Wireshark Packet Capture

WK1-PM4.c — Wireshark PCAP Analysis

Objective:

The objective of this practical was to learn how to capture, analyze, and interpret network traffic using Wireshark. The practical focused on identifying DNS queries and responses, determining IP addresses, analyzing TLS handshakes, identifying Server Name Indication (SNI), and verifying the TLS protocol version used for secure communication.

Wireshark is an open-source network protocol analyzer used to capture and inspect network packets in real time or from previously saved capture files. It helps users understand network communication, troubleshoot connectivity issues, and analyze protocols such as DNS, TCP, and TLS.

Introduction:

In this practical, live network traffic was captured while accessing a website, and a provided PCAP file was analyzed to investigate Microsoft DNS and TLS traffic. Display filters were used to identify relevant packets and examine their details.

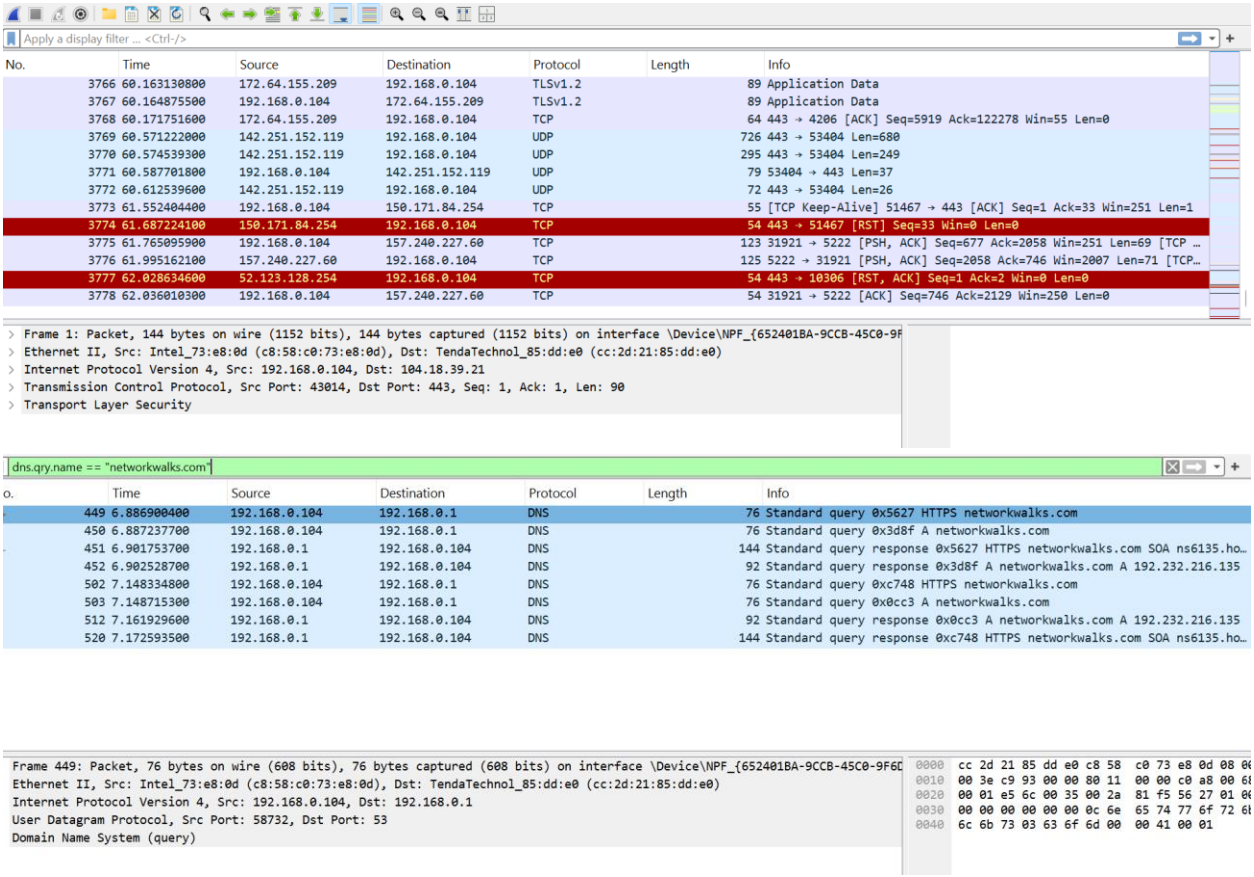
Tools & Technology Used:

- Wireshark Network Protocol Analyzer
- Live network packet capture
- PCAP/PCAPNG capture files
- DNS protocol analysis

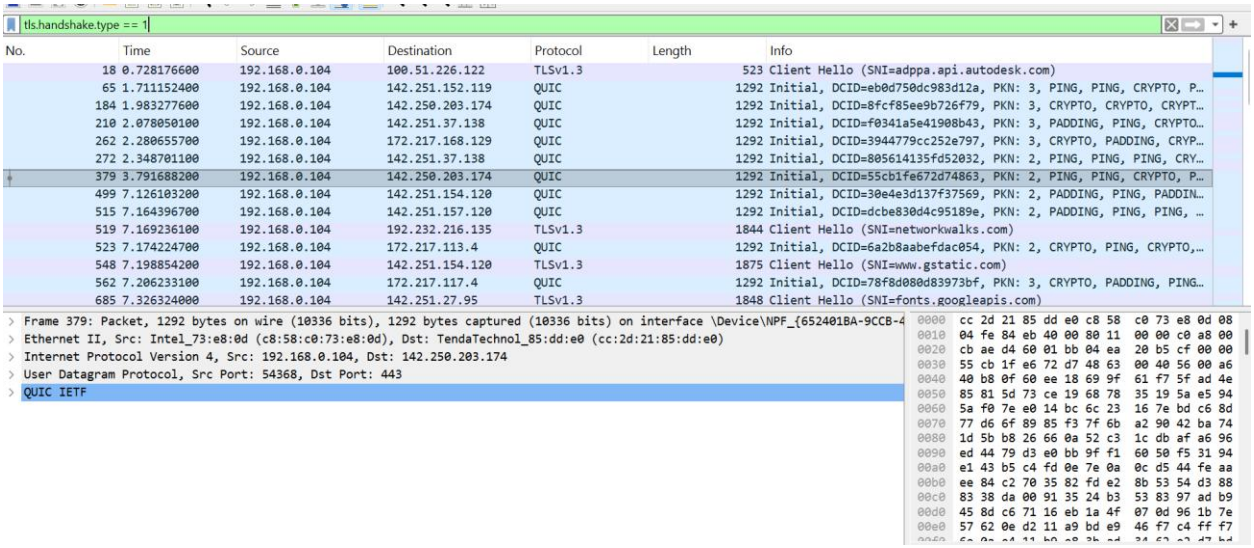
- TCP and TLS protocol analysis
- Wireshark display filters

WK1-PM4.b — Wireshark Packet Capture

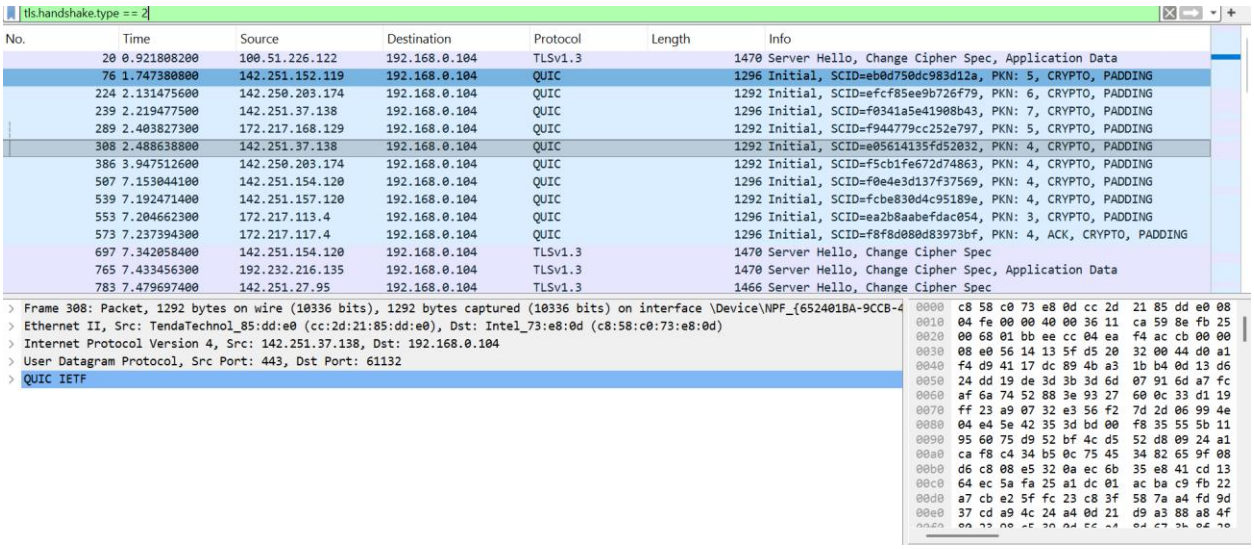
Wireshark DNS Query and Response Analysis for networkwalks.com:



TLS Client Hello and Server Name Indication (SNI) Analysis:



TLS Server Hello and TLS Version Verification for networkwalks.com:



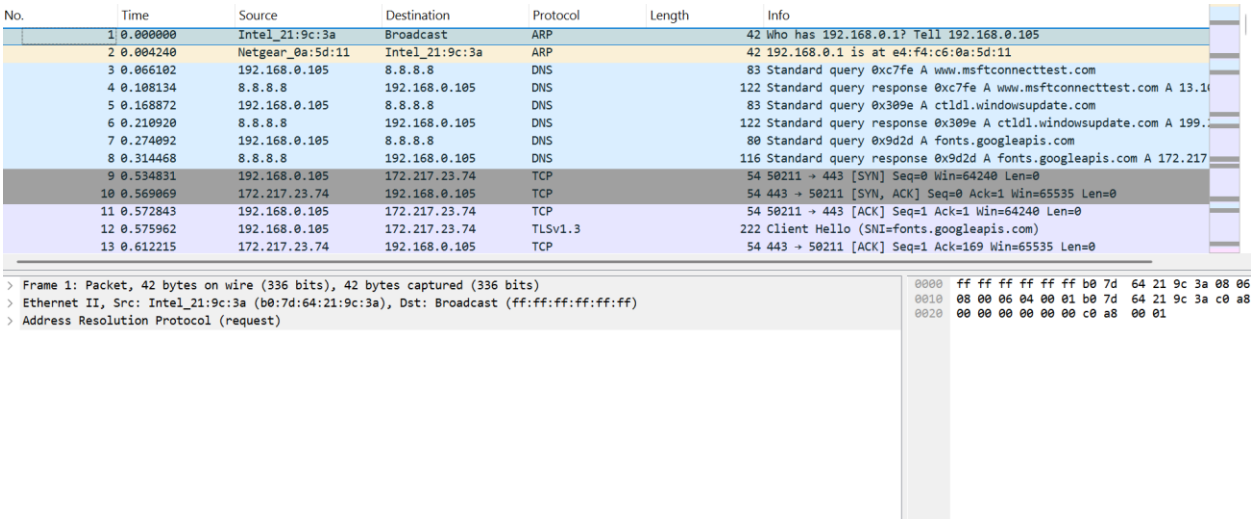
QUESTIONS	ANSWERS
Laptop IP address	192.168.0.104
DNS server IP address	192.168.0.1
Website IP address	192.232.216.135
TLS Client Hello — Server Name	networkwalks.com
Secure server port	443
TLS version	TLSv1.3

Conclusion Lab 01:

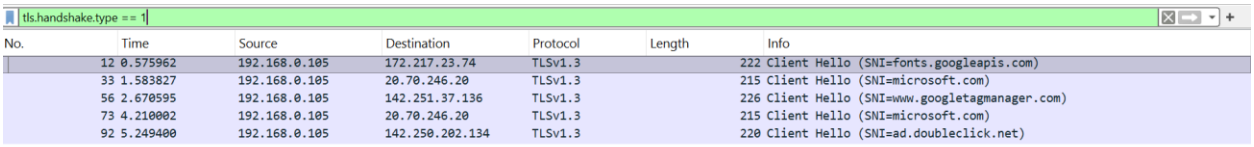
The first practical was completed successfully using Wireshark's live packet capture feature. DNS traffic for networkwalks.com was analyzed to identify the laptop IP address, DNS server IP address, and website IP address. TLS handshake packets were also examined to identify the Server Name Indication (SNI), secure server port, and TLS version. This practical improved understanding of how DNS resolution and secure HTTPS communication work.

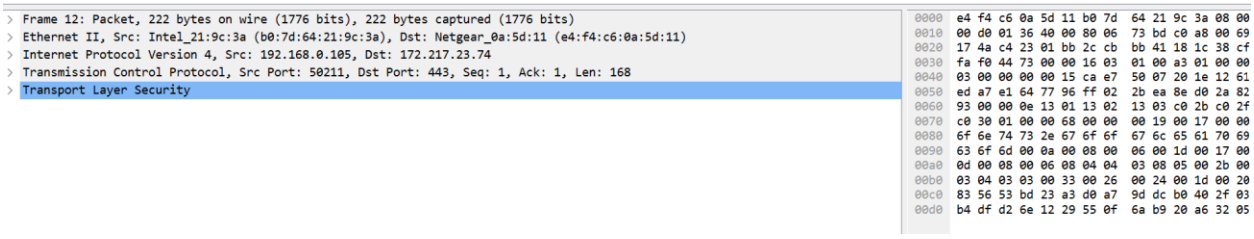
WK1-PM4.c — Wireshark PCAP Analysis

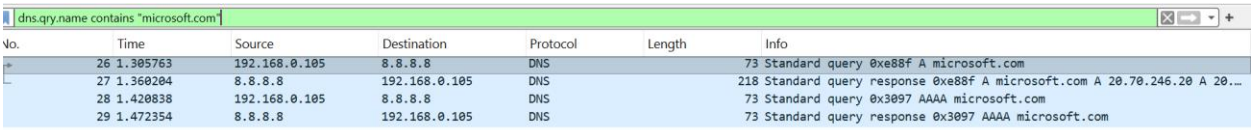
DNS Query and Response Analysis for microsoft.com:

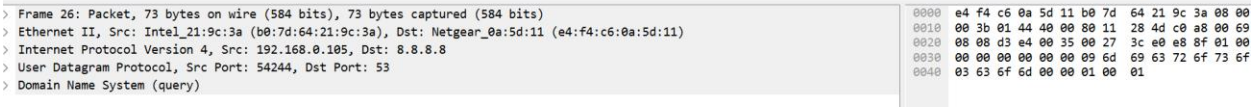


TLS Client Hello and Server Name Indication (SNI) Analysis:



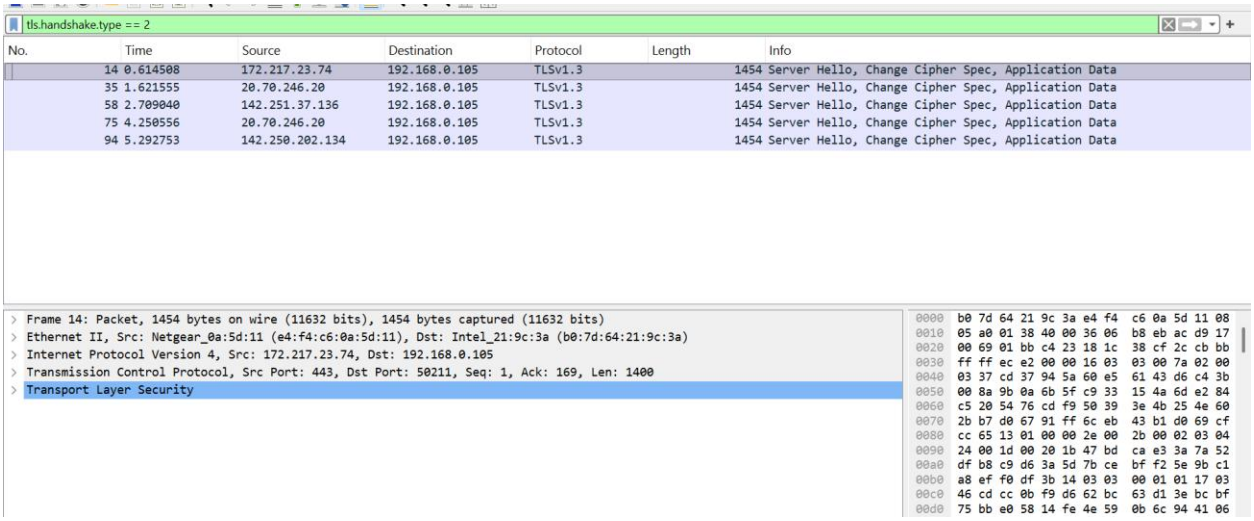






QUESTIONS	ANSWERS
Laptop IP address	192.168.0.105
DNS server IP address	8.8.8.8
Microsoft.com IPv4 addresses	20.70.246.20 , 20.76.201.171 20.112.250.133 , 20.231.239.246 20.236.44.162
TLS Client Hello — Server Name (SNI)	microsoft.com
Secure server port	443
TLS version	TLSv1.3

TLS Server Hello and TLS Version Verification:



Conclusion Lab 02:

The second practical was completed by analyzing the provided Microsoft DNS PCAP file. DNS query and response packets were examined to identify the client IP address, DNS server, and IPv4 addresses associated with microsoft.com. TLS handshake packets were then inspected to identify the SNI, server port, and TLS version. The analysis demonstrated how Wireshark filters can be used to investigate saved network traffic and understand DNS resolution and encrypted communication.

Overall Learning Outcomes:

After completing both practicals, I learned how to capture live network traffic, analyze saved PCAP files, apply Wireshark display filters, identify DNS queries and responses, inspect IP addresses, and examine TLS handshake information. These skills are useful for network troubleshooting, traffic analysis, and basic cybersecurity investigations.